

In-Processing Constraints (Model Fixes)

This step addresses fairness issues inside the model training process itself. It is used when bias cannot be sufficiently reduced through data-level corrections alone. At this stage, fairness is treated as an explicit system requirement, not an indirect outcome. This stage applies when bias originates or persists at the model optimization stage.

This stage must only be applied when causal fairness analysis identifies bias at the model stage or when disparities persist after data-level corrections.

When to Use

Apply the In-Processing stage when:

- Pre-processing data fixes were applied, but unfair outcomes remain
- The model learns or amplifies existing societal or historical biases
- Fairness requirements must be formally enforced during training, rather than assumed through balanced data

This step is especially important for high-impact systems where unequal errors can cause real-world harm.

When Not to Use

In-processing should not be applied when:

- Bias has already been resolved through data-level corrections
- The model cannot be modified (e.g., black-box systems)
- Fairness constraints significantly degrade performance beyond acceptable thresholds

In such cases, alternative intervention strategies must be considered.

Allowed Actions

Teams may implement the following interventions:

- Introduce fairness constraints during model training to limit outcome disparities between protected or impacted groups
- Penalize large errors or performance gaps so the model is discouraged from favoring one group over another
- Reduce reliance on proxy features that indirectly encode sensitive attributes, even if those attributes are not explicitly included

These actions must be applied thoughtfully and tested for stability.

Required Output

All in-processing interventions must be documented with:

- A Model Constraint Summary explaining what constraints were added and the rationale behind them
- A Trade-off Explanation describing impacts on accuracy, performance, or other business objectives

If fairness constraints conflict with system goals, regulatory expectations, or stakeholder priorities, the issue must be escalated for governance review, not resolved through informal or ad-hoc decisions. In black-box systems, this stage may be unavailable and must be formally documented as constrained.

Connection to Intervention Selection and Validation

This stage is selected when causal analysis identifies bias at the model stage or when disparities persist after pre-processing.

All in-processing interventions must be validated to ensure:

- Reduction in disparity metrics
- Acceptable performance trade-offs
- Stability across groups

If validation fails, the system must return to causal analysis or alternative intervention strategies.